
COMPSCI 602

Project Report 5

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Project Title: Research Design for Testing the Causal Role of Late-Layer Refusal-Signal Shifts under Prefilling Attacks

1 Research Questions and Hypotheses

The earlier reports for this project began with a broad question about shallow safety alignment in safety-aligned large language models, then progressively narrowed the focus to a more specific mechanism. Prior work shows that safety alignment in current chat models can be “shallow”: models often refuse harmful prompts initially, but can be pushed toward compliance after only a few forced compliant tokens [2]. Related mechanistic work suggests that refusal behavior is mediated by a sparse internal signal, including a strong linear “refusal direction” in the residual stream [1]. In my exploratory Report 3, the strongest unexpected finding was not simply a smooth positional decay of that signal, but a strong shift toward negative refusal-direction projection in late layers under prefilling attacks, especially at layers 24–27. A small patching pilot at layer 27 did not restore refusal under one intervention setting, so the next step is to design a more systematic study of whether this late-layer shift is causally important.

The single main research question for this report is therefore:

Research Question. *How and under what intervention conditions does the late-layer negative shift in the refusal-direction signal contribute to refusal failure under prefilling attacks?*

This is primarily a *mechanistic* research question. It is not only asking whether refusal decreases under attack; that is already known behaviorally. Instead, it asks how a specific internal representation changes and whether manipulating that representation changes the model’s behavior. It also contains a causal component, because the question is about the effect of targeted interventions on later refusal behavior.

To answer this question efficiently, I will use multiple working hypotheses rather than commit to a single favored explanation. The purpose is not to prove one mechanism correct in advance, but to design evidence that can distinguish among plausible alternatives.

Hypothesis 1: Information-Loss Hypothesis

Under prefilling attacks, refusal fails because the refusal-direction component in late layers is weakened, displaced, or overwritten as generation proceeds. If this hypothesis is correct, then restoring the refusal-direction component at those late attacked positions should increase refusal behavior.

Predictions.

- Harmful prompts under prefilling with $k = 3$ will show a substantially smaller or more negative refusal-direction projection than the baseline $k = 0$ condition in late layers 24 and 27.
- Patching the refusal-direction component from an early or pre-generation source position into attacked late-layer target positions will increase refusal rate relative to the attacked no-patch baseline.

Falsification criterion. If late-layer patching restores little or no refusal even when applied at the layers and token positions showing the largest observed shift, then this hypothesis is weakened.

Hypothesis 2: Active-Suppression Hypothesis

Under prefilling attacks, refusal does not fail merely because the refusal-direction component is lost. Instead, later computations actively counteract or suppress that component. On this view, restoring the direction alone will often be insufficient to recover refusal.

Predictions.

- Late-layer refusal-direction projection under attack will become strongly negative rather than merely shrinking toward zero.
- Patching the refusal-direction component alone will have small or inconsistent behavioral effects, even when targeted to the most affected late layers.

Falsification criterion. If a modest directional intervention in layers 24 or 27 reliably restores refusal across prompts, then active suppression is less plausible as the primary explanation.

Hypothesis 3: Layer-Specificity Hypothesis

If the candidate mechanism is localized, then interventions should matter more in the late layers that exhibited the strongest effect in Report 3 than in earlier comparison layers.

Predictions.

- The attacked-vs.-baseline difference in refusal-direction projection will be much larger in layers 24 and 27 than in layers 16 and 20.
- Patching at layers 24 and 27 will have larger behavioral effects than patching at layer 16.

Falsification criterion. If earlier-layer interventions are equally effective, or if the attacked-vs.-baseline difference is uniform across layers, then the mechanism is less clearly localized to late layers.

Hypothesis 4: Threshold Prefilling Hypothesis

The behavioral and mechanistic effect of prefilling may operate more like a threshold than a monotonic dose-response curve. Report 3 suggested that $k = 3$ already captures most of the attack effect, while $k = 10$ may add little beyond that.

Predictions.

- The main difference will be between $k = 0$ and $k = 3$.
- The difference between $k = 3$ and $k = 10$ will be small relative to the baseline-vs.-attack difference.

Falsification criterion. If larger prefills continue to produce large additional behavioral or mechanistic shifts, then the threshold view is not supported.

Together, these hypotheses define a manageable study. The design is not trying to answer every possible question about shallow alignment. It is designed to determine whether one specific late-layer mechanism identified in the exploratory report is merely correlated with behavioral failure or is plausibly part of the causal process that produces it.

2 Type of Empirical Analysis

This study will use a **mixed empirical design** that combines two sources of evidence:

1. **Observational analysis of internal activations.** I will trace the refusal-direction projection across layers and token positions under baseline and attacked conditions. This part of the study is observational because I am measuring internal behavior without intervening on the model state. The goal is descriptive and associational: characterize where the refusal-direction signal changes, how large those changes are, and how strongly they co-occur with refusal vs. compliance.
2. **Controlled intervention experiments using activation patching.** I will then manipulate the model’s internal state by patching the refusal-direction component into selected late-layer attacked positions. This part of the study is experimental because the intervention is externally imposed and is intended to isolate the effect of a specific internal component on later behavior.

This combination is appropriate for the research question. Pure observation would be insufficient because the refusal-direction shift could be correlated with refusal failure without causing it. Pure intervention without prior tracing would also be weak, because it would not identify where the most informative manipulations should be applied. The observational analysis therefore helps localize the candidate mechanism, and the intervention experiment addresses internal validity by testing whether that localized mechanism is causally important.

I do not plan to use simulation or proof as the primary evidence sources in this report. A formal theoretical account of autoregressive safety decay would be interesting, but the main unresolved issue here is empirical: what happens inside the model under prefilling, and what happens if that internal state is deliberately altered? The most useful evidence for this stage of the project therefore comes from tracing and controlled intervention.

3 Data collection

This section describes the units, treatments, outcomes, settings, and variables that will be measured.

System, task, and environment

System. The primary system will be Llama-3.1-8B-Instruct, the same safety-aligned model used in the earlier reports. Llama-3.2-3B-Instruct may be used only for debugging or smoke tests, not for the main reported analysis.

Task. The task is safety-constrained response generation: given a harmful prompt, the model should refuse to comply.

Environment. The main environment is a prefilling attack in which the model is forced to begin with a compliant prefix before free generation. The main attack condition will use $k = 3$, with $k = 0$ as baseline and $k = 10$ as an optional supplementary comparison.

Units

The primary units of analysis will be **prompt instances**. For the observational tracing analysis, each prompt contributes measurements across multiple conditions, layers, and token positions. For the behavioral analysis, each prompt contributes one model output per condition. For the intervention study, each prompt contributes one output for each patching setting.

The main prompt sets will be:

- **Harmful prompts:** 25 prompts from AdvBench [4].
- **Benign controls:** 25 instruction-following prompts from Alpaca [3].

This is intentionally a modest sample. The main goal of Report 6 is not a giant benchmark, but a careful first study of one mechanism. If the runs are stable and the infrastructure remains reliable, I may later expand the harmful prompt set, but the initial design assumes 25 harmful and 25 benign prompts so that the core experiment is tractable.

Treatments and settings

The main treatments are the controlled variations applied to the model or its internal state.

Prefilling treatment.

- $k = 0$ (baseline, no prefilling)
- $k = 3$ (main attack condition)
- $k = 10$ (supplementary attack condition if time permits)

Layer treatment. For tracing, I will focus on layers {16, 20, 24, 27}. Layers 24 and 27 are the main target layers because Report 3 showed the strongest attacked-vs.-baseline shift there. Layers 16 and 20 will serve as comparison layers.

Patching treatment. For the intervention experiment, I will patch the refusal-direction component into selected attacked target positions using a small factorial design rather than an exhaustive sweep. The main manipulated factors will be:

- target layer: {16, 24, 27}
- source position: last input/prefill token and one early generated token
- target token position: generated positions {1, 3, 5}

Rationale for selected conditions

The design intentionally focuses on a small set of conditions rather than a large factorial sweep. The choice of $k = 3$ as the main attacked condition is motivated by the exploratory findings from Report 3, where $k = 3$ already produced the main behavioral drop and a strong late-layer negative shift, while $k = 10$ added little additional change. Layers 24 and 27 are the primary intervention targets because they showed the strongest negative shift in the exploratory tracing results, while layers 16 and 20 serve as comparison layers to test whether any patching effect is specific to the late layers rather than a generic intervention effect. Target token positions {1, 3, 5} are chosen to cover the early part of free generation, where the refusal-to-compliance transition is most likely to emerge while still keeping the intervention space manageable. The last input or prefill token is used as a primary source position because it is the final shared context state before free generation begins, making it a natural candidate source for restoring the refusal-direction component. This narrow design follows the course recommendation to use a small number of discrete factor levels, identify main effects first, and avoid unnecessarily large factorial designs.

Settings held constant.

- model weights
- tokenizer and chat template
- decoding configuration
- refusal-direction extraction procedure
- maximum generation length
- prompt set within each analysis

Holding these settings fixed is important because the study is already varying several factors. Introducing unnecessary variation would make the results harder to interpret.

Outcomes and measured variables

The measured outcomes fall into two groups: mechanistic variables and behavioral variables.

Mechanistic variables.

- **Refusal-direction projection:** scalar projection of the residual stream activation onto the extracted refusal direction.
- **Layer index:** the transformer layer at which the projection is measured.

- **Generated token position:** the position in the generated continuation at which the projection is measured.

Behavioral variables.

- **Refusal label:** binary refusal vs. non-refusal outcome assigned using the current phrase-based classifier.
- **Patch effect:** change in refusal behavior between the attacked no-patch condition and the attacked patched condition.

Because the refusal classifier is relatively simple, I will also manually inspect a small number of borderline outputs if the automatic label appears questionable. This is mainly a construct-validity safeguard, not a separate evaluation method.

Units of measurement

- refusal-direction projection: real-valued scalar
- refusal label: binary indicator (0/1)
- refusal rate: proportion of prompts labeled refusal
- patch effect: difference in refusal rate or difference in per-prompt refusal label before vs. after intervention

The observational tracing dataset will therefore consist of $\text{prompt} \times \text{condition} \times \text{layer} \times \text{token-position}$ measurements. The interventional dataset will consist of $\text{prompt} \times \text{patch-setting}$ outputs.

4 Analytic techniques

The analysis will emphasize clear visualization, paired comparisons, and simple summary statistics rather than heavy modeling. This is appropriate because the study is focused on one mechanism, the sample size is moderate, and interpretability is more important than methodological complexity.

Mapping hypotheses to empirical analyses

Analysis for Hypothesis 1 (information-loss hypothesis). I will compare refusal-direction projection between $k = 0$ and $k = 3$ for each layer and generated token position. The main analyses will be heatmaps and layer-wise line plots over token position, together with summary tables of mean projection by layer and condition. Support for Hypothesis 1 would come from a clearly smaller or more negative late-layer projection under prefilling, followed by a measurable increase in refusal after late-layer patching. If late-layer patching leaves refusal essentially unchanged even at the most affected positions, the information-loss account is weakened.

Analysis for Hypothesis 2 (active-suppression hypothesis). I will examine whether the attacked late-layer projection becomes strongly negative rather than simply shrinking toward zero, and whether that negative shift persists even when directional patching is applied. The main analyses will use prompt-level raw-data plots for layer 27, before/after patching tables, and effect-size summaries for attacked no-patch versus patched conditions. Support for Hypothesis 2 would come from a strong negative late-layer shift together with weak or inconsistent restoration from directional patching alone.

Analysis for Hypothesis 3 (layer-specificity hypothesis). I will compare attacked-vs.-baseline projection differences across layers 16, 20, 24, and 27, and compare behavioral patch effects across target layers 16, 24, and 27. The main outputs will be per-layer mean-difference tables, paired prompt-level comparisons, and small-factor comparison plots. Support for Hypothesis 3 would come from substantially larger mechanistic and behavioral effects in layers 24 and 27 than in layer 16.

Analysis for Hypothesis 4 (threshold-prefilling hypothesis). I will compare behavioral and mechanistic differences across $k = 0$, $k = 3$, and, if available, $k = 10$. The main outputs will be refusal-rate tables and summary plots of projection by condition. Support for Hypothesis 4 would come from a large baseline-vs.-attack difference but only a small additional shift from $k = 3$ to $k = 10$.

Behavioral and uncertainty summaries

For behavioral outputs, I will use refusal-rate tables for baseline, attacked, and patched conditions, paired prompt-level comparisons between conditions, and confidence intervals for refusal-rate differences. Because the sample is modest, I will likely use bootstrap intervals or exact binomial intervals rather than rely on asymptotic approximations.

If the patching data are clean enough, I may also apply a simple paired binary test such as McNemar’s test for attacked no-patch vs. patched outcomes. However, I will not force formal significance testing if the number of discordant cases is too small to make it informative. In that case, I will report effect sizes and uncertainty intervals instead.

Interpreting outcomes

The interpretation of the study is straightforward:

- Strong late-layer negative shifts plus successful restoration through patching would support the information-loss account.
- Strong late-layer negative shifts plus weak or null restoration would support the active-suppression account, or at least indicate that the refusal direction alone is not sufficient.
- Larger effects in layers 24 and 27 than in layer 16 would support the localization hypothesis.
- Similar outcomes for $k = 3$ and $k = 10$ would support the threshold-prefilling hypothesis.

Overall, this design is intentionally narrow. It does not attempt to explain all failure modes of aligned language models, compare all attack families, or trace every possible internal substrate. Its purpose is to obtain a credible, interpretable answer to one mechanistic question that emerged directly from the earlier exploratory results.

References

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